**TASK 3- Titanic Survival Prediction using Machine Learning**

This project focuses on building a machine learning classification model to predict passenger survival on the Titanic using passenger-related attributes. The project emphasizes feature engineering, preprocessing, model training, and model explainability.

Exploratory Data Analysis (EDA) was performed to analyze dataset structure, missing values, and feature distributions. Several feature engineering techniques were implemented including extracting passenger titles from names, creating a FamilySize feature, and identifying cabin presence.

Missing values in the Age and Embarked columns were handled using statistical imputation techniques. Categorical variables were encoded using Label Encoding for machine learning compatibility.

Two machine learning classification algorithms were trained and compared:

1. Logistic Regression
2. Random Forest Classifier

The models were evaluated using accuracy score and classification reports. Feature importance analysis was performed using Random Forest to understand which features contributed most to survival prediction.

Random Forest achieved the best performance and was selected as the final model. The trained model was saved using Joblib and tested with example inference input.

This project demonstrates a complete beginner-level machine learning classification workflow including preprocessing, feature engineering, model evaluation, explainability, and model persistence.